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GATE ECE · ELECTRONIC DEVICES AND VLSI · CHAPTER-WISE PYQS

Semiconductor Physics

GATE ECE Electronic Devices and VLSI · Previous Year Questions with Answer Key

56	28	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	34 questions
Marks 2	22 questions
Answer key	at the end

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Marks 1 34 questions

Q1. GATE ECE 2025

Which of the following can be used as an n-type dopant for silicon?

Select the correct option(s).

(A) Arsenic (B) Boron (C) Gallium (D) Phosphorous

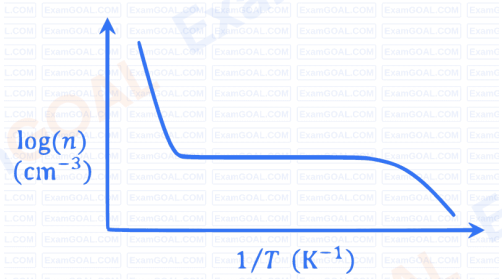
Open solution

Q2. GATE ECE 2024

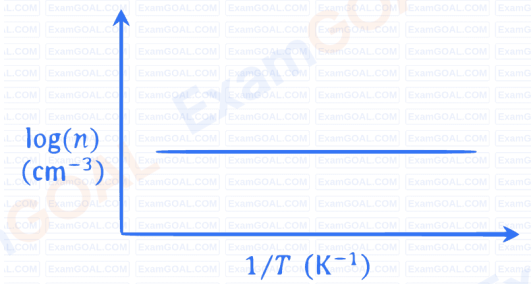
For non-degenerately doped n-type silicon, which one of the following plots represents the temperature (T) dependence of free electron concentration (n)?

Open solution

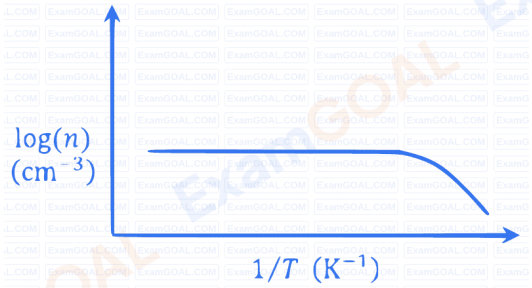
(A)



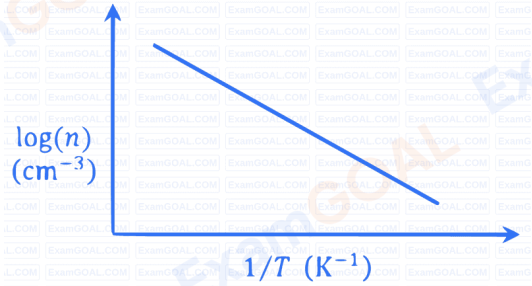
(B)



(C)



(D)

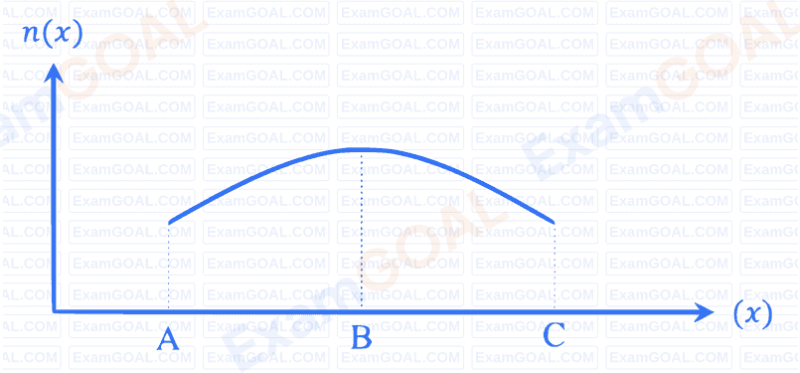


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Q3. GATE ECE 2024

The free electron concentration profile $n(x)$ in a doped semiconductor at equilibrium is shown in the figure, where the points A, B, and C mark three different positions. Which of the following statements is/are true?

Open solution



(A) For x between B and C, the electron diffusion current is directed from C to B.

(B) For x between B and A, the electron drift current is directed from B to A.

(C) For x between B and C, the electric field is directed from B to C.

(D) For x between B and A, the electric field is directed from A to B.

Q4. GATE ECE 2023

In a semiconductor, if the Fermi energy level lies in the conduction band, then the semiconductor is known as

Open solution

(A) degenerate n-type.

(B) degenerate p-type.

(C) non-degenerate n-type.

(D) non-degenerate p-type.

Q5. GATE ECE 2023

For an intrinsic semiconductor at temperature $T = 0K$, which of the following statement is true?

Open solution

(A) All energy states in the valence band are filled with electrons and all energy states in the conduction band are empty of electrons.

(B) All energy states in the valence band are empty of electrons and all energy states in the conduction band are filled with electrons.

(C) All energy states in the valence and conduction band are filled with holes.

(D) All energy states in the valence and conduction band are filled with electrons.

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Q6. GATE ECE 2022

Open solution

Consider a long rectangular bar of direct bandgap p-type semiconductor. The equilibrium hole density is 10^{17} cm^{-3} and the intrinsic carrier concentration is 10^{10} cm^{-3}. Electron and hole diffusion lengthss are 2 μm and 1 μm, respectively. The left side of the bar (x = 0) is uniformly illuminated with a laser having photon energy greater than the bandgap of the semiconductor. Excess electron-hole pairs are generated ONLY at x = 0 because of the laser. The steady state electron density at x = 0 is 10^{14} cm^{-3} due to laser illumination. Under these conditions and ignoring electric field, the closest approximation (among the given options) of the steady state electron density at x = 2 μm, is _____.

(A) 0.37×10^{14} cm^{-3}
(B) 0.63×10^{13} cm^{-3}
(C) 3.7×10^{14} cm^{-3}
(D) 10^{3} cm^{-3}

Q7. GATE ECE 2022

Open solution

In a non-degenerate bulk semiconductor with electron density n = 10^{16} cm^{-3}, the value of $E_C - E_{Fn}$ = 200 meV, where E_C and E_{Fn} denote the bottom of the conduction band energy and electron Fermi level energy, respectively. Assume thermal voltage as 26 meV and the intrinsic carrier concentration is 10^{10} cm^{-3}. For n = 0.5×10^{16} cm^{-3}, the closest approximation of the value of $(E_C - E_{Fn})$, among the given options is _____.

(A) 226 meV
(B) 174 meV
(C) 218 meV
(D) 182 meV

Q8. GATE ECE 2020

Open solution

A single crystal intrinsic semiconductor is at a temperature of 300 K with effective density of states for holes twice that of electrons. The thermal voltage is 26 mV . The intrinsic Fermi level is shifted from midbandgap energy level by

(A) 9.01 meV
(B) 13.45 meV
(C) 18.02 meV
(D) 26.90 meV

Q9. GATE ECE 2017 Set 1

Open solution

A bar of Gallium Arsenide (GaAs) is doped with Silicon such that the Silicon atoms occupy Gallium and Arsenic sites in the GaAs crystal. Which one of the following statement is true?

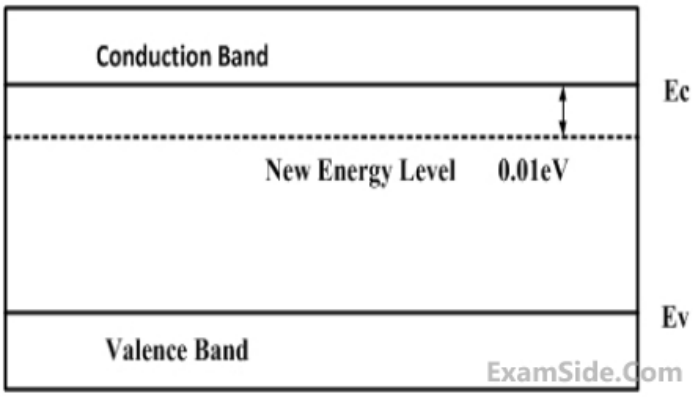
(A) Silicon atoms act as p-type dopants in Arsenic sites and n-type dopants in Gallium sites
(B) Silicon atoms act as n-type dopants in Arsenic sites and p-type dopants in Gallium sites
(C) Silicon atoms act as p-type dopants in Arsenic as well as Gallium sites
(D) Silicon atoms act as n-type dopants in Arsenic as well as Gallium sites

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Q10. GATE ECE 2016 Set 1

Open solution

A small percentage of impurity is added to an intrinsic semiconductor at 300 K. Which one of the following statements is true for the energy band diagram shown in the following figure?



(A) Intrinsic semiconductor doped with pentavalent atoms to form n-type semiconductor
(B) Intrinsic semiconductor doped with trivalent atoms to form n-type semiconductor
(C) Intrinsic semiconductor doped with pentavalent atoms to form p-type semiconductor
(D) Intrinsic semiconductor doped with trivalent atoms to form p-type semiconductor

Q11. GATE ECE 2015 Set 2

Open solution

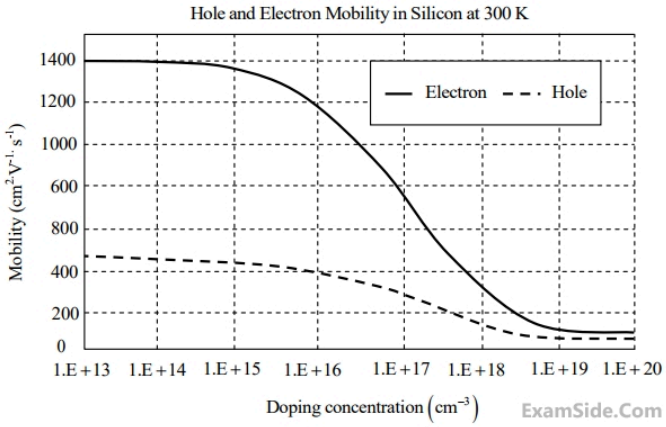
An n–type silicon sample is uniformly illuminated with light which generates 10^{20} electron hole pairs per cm^3 per second. The minority carrier lifetime in the sample is 1 μs.In the steady state, the hole concentration in the sample is approximately 10^x , where x is an integer. The value of x is _____

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Q12. GATE ECE 2015 Set 2

Open solution

A piece of silicon is doped uniformly with phosphorous with a doping concentration of $10^{16}/cm^2$. The expected value of mobility versus doping concentration for silicon assuming full dopant ionization is shown below. The charge of an electron is $1.6 \times 10^{-19}\ C$. The conductivity (in S cm^{-1}) of the silicon sample at 300 K is _____.



Q13. GATE ECE 2015 Set 1

Open solution

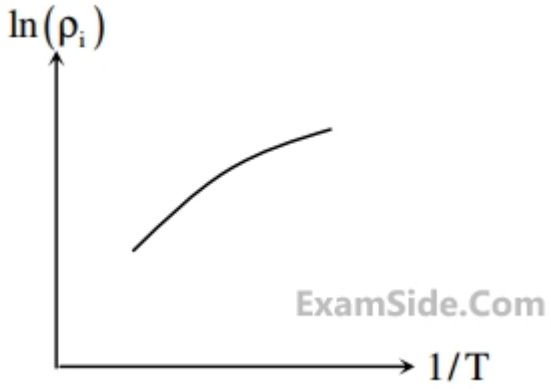
A silicon sample is uniformly doped with donor type impurities with a concentration of $10^{16}/cm^3$.The electron and hole mobilities in the sample are $1200\ cm^2/V-s$ and $400\ cm^2/V-s$ respectively. Assume complete ionization of impurities. The charge of an electron is $1.6 \times 10^{-19}\ C$.The resistivity of the sample $(in\ \Omega - cm)$ is _____.

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Q14. GATE ECE 2014 Set 4

In the figure ln(ρ_i) is plotted as a function of 1/T, where ρ_i the intrinsic resistivity of silicon, T is is the temperature, and the plot is almost linear.

Open solution



The slope of the line can be used to estimate

(A) Band gap energy of silicon (E_g)

(B) Sum of electron and hole mobility in silicon $(\mu_n + \mu_p)$

(C) Reciprocal of the sum of electron and hole mobility in silicon $(\mu_n + \mu_p)^{-1}$

(D) Intrinsic carrier concentration of silicon (n_i)

Q15. GATE ECE 2014 Set 4

The cut-off wavelength (in μm) of light that can be used for intrinsic excitation of a semiconductor material of bandgap E_g= 1.1 eV is _____.

Open solution

Q16. GATE ECE 2014 Set 3

A thin P-type silicon sample is uniformly illuminated with light which generates excess carriers. The recombination rate is directly proportional to

Open solution

(A) The minority carrier mobility

(B) The minority carrier recombination lifetime

(C) The majority carrier concentration

(D) The excess minority carrier concentration

Q17. GATE ECE 2014 Set 3

At T = 300 K, the hole mobility of a semiconductor $\mu_p = 500\ cm^2/V-s$ and $\frac{kT}{q} = 26\ mV$.The hole diffusion constant D_p in cm^2/s is_____.

Open solution

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Q18. GATE ECE 2011

Drift current in the semiconductors depends upon

(A) only the electric field

(B) only the carrier concentration gradient

(C) both the electric field and the carrier concentration

(D) both the electric field and the carrier concentration gradient

Open solution

Q19. GATE ECE 2008

Which of the following is true?

(A) A silicon wafer heavily doped with boron is a p^+ substrate

(B) A silicon wafer lightly doped with boron is a p^+ substrate

(C) A silicon wafer heavily doped with arsenic is a p^+ substrate

(D) A silicon wafer lightly doped with arsenic is a p^+ substrate

Open solution

Q20. GATE ECE 2006

Under low level injection assumption, the injected minority carrier current for an extrinsic semiconductor is essentially the

(A) diffusion current

(B) drift current

(C) recombination current

(D) induced current

Open solution

Q21. GATE ECE 2006

The concentration of minority carriers in an extrinsic semiconductor under equilibrium is:

(A) directly proportional to the doping concentration

(B) inversely proportional to the doping concentration

(C) directly proportional to the intrinsic concentration

(D) inversely proportional to the intrinsic concentration

Open solution

Q22. GATE ECE 2005

The primary reason for the widespread use of Silicon in semiconductor device technology is

(A) abundance of Silicon on the surface of the Earth.

(B) larger bandgap of Silicon in comparison to Germanium.

(C) favorable properties of Silicon-dioxide (SiO_2)

(D) lower melting point

Open solution

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Q23. GATE ECE 2005

A Silicon PN junction at a temperature of 20°C has a reverse saturation current of 10 pico-Amperes (pA). The reverse saturation current at 40°C for the same bias is approximately

(A) 20 pA (B) 30 pA (C) 40 pA (D) 80 pA

Open solution

Q24. GATE ECE 2005

The band gap of Silicon at room temperature is:

(A) 1.3 eV (B) 0.7 eV (C) 1.1 eV (D) 1.4 eV

Open solution

Q25. GATE ECE 2004

The impurity commonly used for realizing the base region of a silicon n-p-n transistor is

(A) Gallium (B) Indium (C) Boron (D) Phosphorus

Open solution

Q26. GATE ECE 2003

n-type silicon is obtained by doping silicon with

(A) Germanium (B) Aluminum (C) Boron (D) Phosphorus

Open solution

Q27. GATE ECE 2002

The band gap of silicon at 300 K is

(A) 1.36 eV (B) 1.10 eV (C) 0.80 eV (D) 0.67 eV

Open solution

Q28. GATE ECE 2002

The intrinsic carrier concentration of silicon sample at 300°K is $1.5 \times 10^{16}/m^3$. If after doping, the number of majority carriers is $5 \times 10^{20}/m^3$, minority carrier density is

(A) $4.50 \times 10^{11}/m^3$ (B) $3.33 \times 10^{4}/m^3$ (C) $5.00 \times 10^{20}/m^3$ (D) $3.00 \times 10^{-5}/m^3$

Open solution

Q29. GATE ECE 1997

The units of $\frac{q}{KT}$ are

(A) V (B) V^{-1} (C) J (D) J/K

Open solution

Q30. GATE ECE 1995

The probability that an electron in a metal occupies the Fermi-level at any temperature (> 0 K)

(A) 0 (B) 1 (C) 0.5 (D) 1.0

Open solution

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Q31. GATE ECE 1995

The drift velocity of electrons in silicon

(A) is proportional to the electric field for all values of electric field

(B) is independent of the electric field

(C) increases at low values of electric field and decreases at high values of electric field exhibiting negative differential resistance.

(D) increases linearly with electric field at low values of electric field and gradually saturates at higher values of electric field.

Open solution

Q32. GATE ECE 1995

In a P-type silicon sample, the hole concentration is $2.25 \times 10^{15}/cm^3$. If the intrinsic carrier concentration is $1.5 \times 10^{10}/cm^3$, the electron concentration is

(A) Zero (B) $10^{10}/cm^3$ (C) $10^5/cm^3$ (D) $1.5 \times 10^{25}/cm^3$

Open solution

Q33. GATE ECE 1994

A p-type silicon sample has a higher conductivity compared to an n-type sample having the same dopant concentration.

(A) TRUE (B) FALSE

Open solution

Q34. GATE ECE 1994

A small concentration of minority carries is injected into a homogeneous semiconductor crystal at one point. An electric field of 10 V/cm is applied across the crystal and this moves the minority carriers a distance of 1 cm is 20 μ sec. The mobility (in cm^2/volt-sec) will be

(A) 1,000 (B) 2,000 (C) 5,000 (D) 500,000

Open solution

Marks 2 22 questions

Q35. GATE ECE 2026

Consider that the concentration of electrons in a semiconductor bar varies linearly from $2 \times 10^{17}\ \mathrm{cm}^{-3}$ at $x = 1\mu\ \mathrm{m}$ to $1 \times 10^{16}\ \mathrm{cm}^{-3}$ at $x = 4\mu\ \mathrm{m}$ along the x-direction. Assume that the concentration of electrons is not varying along other directions (that is along y and z-directions).

[Given: the mobility of electron is $1400\ \mathrm{cm}^2\ \mathrm{V}^{-1}\ \mathrm{s}^{-1}$, thermal voltage is 25 mV and electronic charge is 1.6×10^{-19} Coulomb.]

The density of electron diffusion current (in $\mathrm{A/mm}^2$) is ____ .

(rounded off to two decimal places)

Open solution

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Q36. GATE ECE 2025

The intrinsic carrier concentration of a semiconductor is $2.5 \times 10^{16}/\mathrm{m}^3$ at 300 K . If the electron and hole mobilities are $0.15\ \mathrm{m}^2/\mathrm{Vs}$ and $0.05\ \mathrm{m}^2/\mathrm{Vs}$, respectively, then the intrinsic resistivity of the semiconductor (in $\mathrm{k}\Omega.\,\mathrm{m}$) at 300 K is _____ (Charge of an electron $e = 1.6 \times 10^{-19}\mathrm{C}$.)

(A) 1.65 (B) 1.25 (C) 0.85 (D) 1.95

Open solution

Q37. GATE ECE 2025

The electron mobility μ_n in a non-degenerate germanium semiconductor at 300 K is $0.38\ \mathrm{m}^2/\mathrm{Vs}$.

The electron diffusivity D_n at 300 K (in cm^2/s, rounded off to the nearest integer) is _____

(Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23}\ \mathrm{J/K}$ and the charge of an electron $e = 1.6 \times 10^{-19}\mathrm{C}$.)

(A) 26 (B) 98 (C) 38 (D) 10

Open solution

Q38. GATE ECE 2024

A non-degenerate n-type semiconductor has 5 % neutral dopant atoms. Its Fermi level is located at 0.25 eV below the conduction band (E_C) and the donor energy level (E_D) has a degeneracy of 2. Assuming the thermal voltage to be 20 mV, the difference between E_C and E_D (in eV, *rounded off to two decimal places*) is _____.

Open solution

Q39. GATE ECE 2023

In an extrinsic semiconductor, the hole concentration is given to be $1.5n_i$ where n_i is the intrinsic carrier concentration of $1 \times 10^{10}\ cm^{-3}$. The ratio of electron to hole mobility for equal hole and electron drift current is given as _____ (rounded off to two decimal places).

Open solution

Q40. GATE ECE 2023

In a semiconductor device, the Fermi-energy level is 0.35 eV above the valence band energy. The effective density of states in the valence band at T = 300 K is $1 \times 10^{19}\ \mathrm{cm}^{-3}$. The thermal equilibrium hole concentration in silicon at 400 K is _____ $\times 10^{13}\ \mathrm{cm}^{-3}$ (rounded off to two decimal places).

Given kT at 300 K is 0.026 eV.

Open solution

Q41. GATE ECE 2022

Select the CORRECT statements regarding semiconductor devices

(A) Electrons and holes are of equal density in an intrinsic semiconductor at equilibrium.

(B) Collector region is generally more heavily doped than Base region in a BJT.

(C) Total current is spatially constant in a two terminal electronic device in dark under steady state condition.

(D) Mobility of electrons always increases with temperature in Silicon beyond 300 K.

Open solution

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Q42. GATE ECE 2021

Open solution

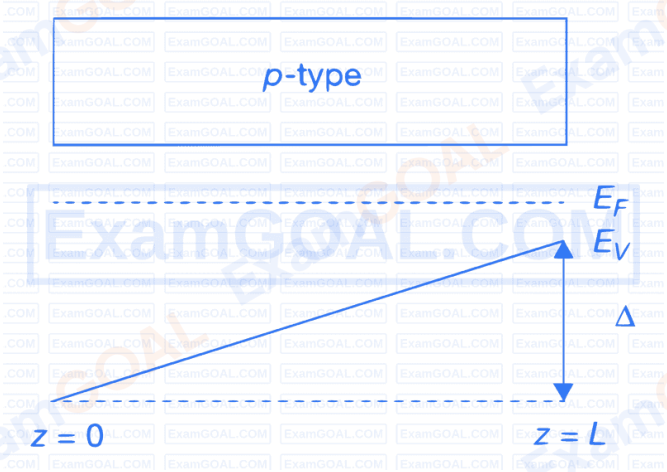
A bar of silicon is doped with boron concentration of 10^{16}cm^{-3} and assumed to be fully ionized. It is exposed to light such that electron-hole pairs are generated throughout the volume of the bar at the rate of $10^{20}\ \text{cm}^{-2}\ \text{s}^{-1}$. If the recombination lifetime is $100\mu\ \text{s}$, intrinsic carrier concentration of silicon is $10^{10}\ \text{cm}^{-3}$ and assuming 100% ionization of boron, then the approximate product of steady-state electron and hole concentrations due to this light exposure is

(A) $10^{20}\ \text{cm}^{-6}$
(B) $2 \times 10^{32}\ \text{cm}^{-6}$
(C) $10^{32}\ \text{cm}^{-6}$
(D) $2 \times 10^{20}\ \text{cm}^{-6}$

Q43. GATE ECE 2021

Open solution

The energy band diagram of a p-type semiconductor bar of length L under equilibrium condition (i.e., the Fermi energy level E_F is constant) is shown in the figure. The valance band E_V is sloped since doping is non-uniform along the bar. The different between the energy levels of the valence band at the two edges of the bar is Δ.



If the charge of an electron is q, then the magnitude of the electric field developed inside the semiconductor bar is

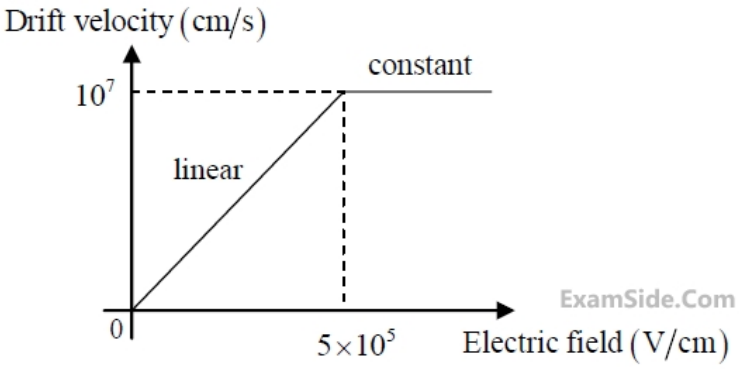
(A) $\frac{2\Delta}{qL}$
(B) $\frac{\Delta}{2qL}$
(C) $\frac{\Delta}{qL}$
(D) $\frac{3\Delta}{2qL}$

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Q44. GATE ECE 2017 Set 1

Open solution

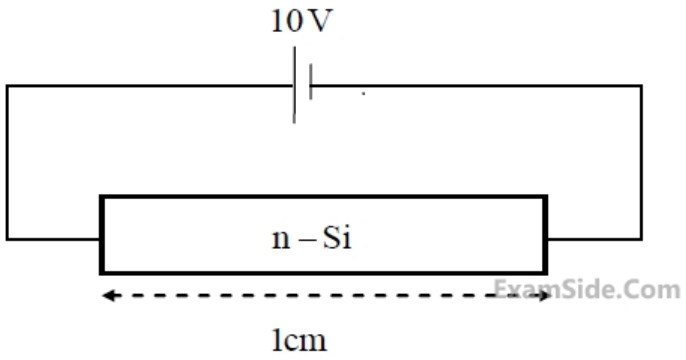
The dependence of drift velocity of electrons on electric field in a semiconductor is shown below. The semiconductor has a uniform electron concentration of n = 1x10^{16} cm^{-3} and electronic charge q = 1.6x10^{-19} C. If a bias of 5V is applied across a 1 μm region of this semiconductor, the resulting current density in this region, in kA/cm^2, is _____.



Q45. GATE ECE 2015 Set 2

Open solution

A dc voltage of 10V is applied across an n–type silicon bar having a rectangular cross–section and a length of 1cm as shown in figure. The donor doping concentration N_D and the mobility of electrons μn are 10^{16} cm^{-3} and 1000 cm^2 V^{-1}s^{-1}, respectively. The average time (in μs) taken by the electrons to move from one end of the bar to other end is _____.



Q46. GATE ECE 2014 Set 4

Open solution

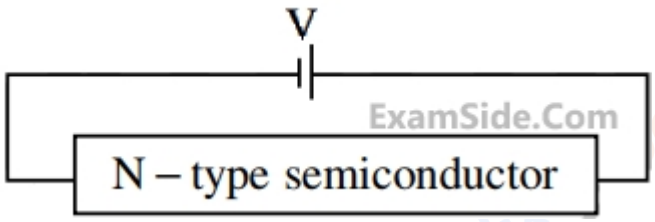
Consider a silicon sample doped with N_D = 1×10^{15}/cm^3 donor atoms. Assume that the intrinsic carrier concentration n_i = 1.5×10^{10}/cm^3. If the sample is additionally doped with N_A = 1×10^{18}/cm^3 acceptor atoms, the approximate number of electrons/cm^3 in the sample, at T=300 K, will be _____.

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Q47. GATE ECE 2014 Set 4

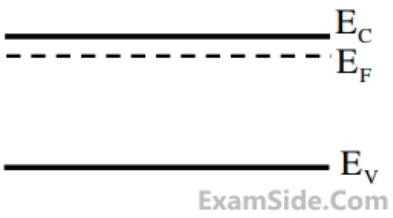
Open solution

An N-type semiconductor having uniform doping is biased as shown in the figure.



If E_C is the lowest energy level of the conduction band, E_V is the highest energy level of the valance band and E_F is the Fermi level, which one of the following represents the energy band diagram for the biased N-type semiconductor?

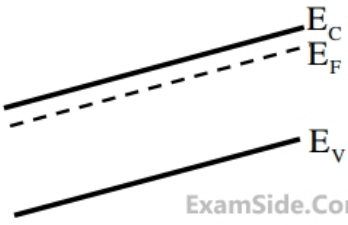
(A)



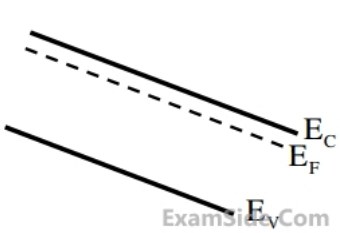
(B)



(C)



(D)



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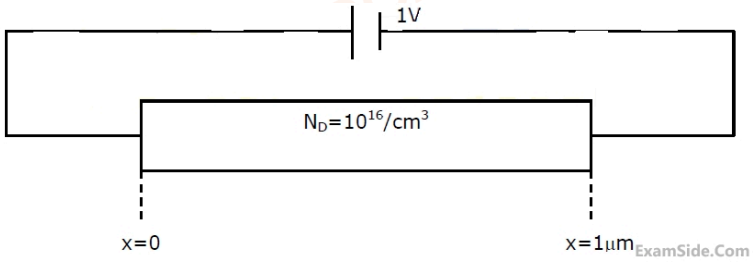
Q48. GATE ECE 2014 Set 2

Assume electronic charge q = 1.6×10^{-19} C, kT/q = 25 mV and electron mobility μ_n = 1000 cm^2/V-s. If the concentration gradient of electrons injected into a P-type silicon sample is $1\times10^{21}/cm^4$, the magnitude of electron diffusion current density (in A/cm^2) is _____.

Open solution

Q49. GATE ECE 2010

The silicon sample with unit cross-sectional area shown below is in thermal equilibrium. The following information is given: T=300K, electronic charge=$1.6x10^{-19}$C, thermal voltage=26mV and electron mobility = $1350cm^2$/V-s



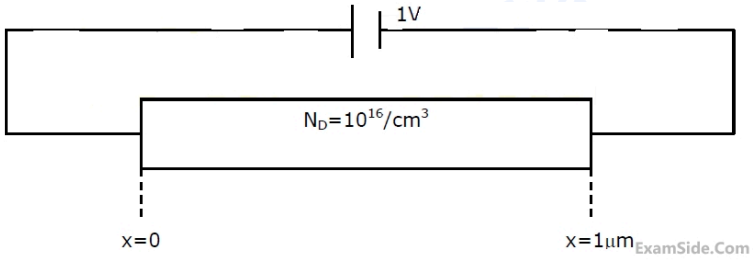
The magnitude of the electric field at x=0.5 μm is

(A) 1 KV/cm (B) 5 KV/cm (C) 10 KV/cm (D) 25 KV/cm

Open solution

Q50. GATE ECE 2010

The silicon sample with unit cross-sectional area shown below is in thermal equilibrium. The following information is given: T=300K, electronic charge=$1.6x10^{-19}$C, thermal voltage=26mV and electron mobility = $1350cm^2$/V-s



The magnitude of the electron drift current density at x=0.5 μm is

(A) $2.16x10^4$ A/cm^2 (B) $1.08x10^4$ A/cm^2 (C) $4.32x10^3$ A/cm^2 (D) $6.48x10^2$ A/cm^2

Open solution

Q51. GATE ECE 2005

A silicon sample 'A' is doped with 10^{18} atoms/cm^3 of Boron. Another sample 'B' of identical dimensions is doped with 10^{18} atoms/cm^3 of Phosphorus. The ratio of electron to hole mobility is 1/3. The ratio of conductivity of the sample A to B is

(A) 3 (B) 1/3 (C) 2/3 (D) 3/2

Open solution

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Q52. GATE ECE 2003

The electron concentration in a sample of uniformly doped n-type silicon at 300°K varies linearly from $10^{17}/cm^3$ at x = 0 to $6 \times 10^{16}/cm^3$ at x = 2 μm. Assume a situation that electrons are supplied to keep this concentration gradient constant with time.If electronic charge is 1.6×10^{-19} $coulomb$ and the diffusion constant $D_n = 35\ cm^2/s$, the current density in the silicon, if no electric field is present is

(A) Zero (B) 120 A/cm^2 (C) +1120 A/cm^2 (D) -1120 A/cm^2

Open solution

Q53. GATE ECE 2003

An n-type silicon bar 0.1 cm long and $100\ \mu m^2$ in cross-sectional area has a majority carrier concentration of $5 \times 10^{20}/m^3$ and the carrier mobility is $0.13\ \ m^2/v-s$ at 300°K. if the charge of an electron is 1.6×10^{-19} coulomb, then the resistance of the bar is

(A) $10^6\ \Omega$ (B) $10^4\ \Omega$ (C) $10^{-1}\ \Omega$ (D) $10^{-4}\ \Omega$

Open solution

Q54. GATE ECE 1992

A semiconductor is irradiated with light such that carriers are uniformly generated throughout its volume. The semiconductor is n-type with $N_D = 10^{19}/cm^3$. If the excess electron concentration in the steady state id $\triangle n = 10^{15}/cm^3$ and if $\tau_p = 10\ \mu\ sec$ [minority carrier life time] the generation rate due to irradiation

(A) $is\ 10^{20}\ e-h\ pairs/cm^3/sec$

(B) $is\ 10^{24}\ e-h\ pairs/cm^3/sec$

(C) $is\ 10^{10}\ e-h\ pairs/cm^3/sec$

(D) cannot be determined as the given data is insufficient

Open solution

Q55. GATE ECE 1991

A silicon sample is uniformly doped with 10^{16} phosphorous atoms/cm^3 and 2 ×10^{16} boron atoms/cm^3. If all the dopants are fully ionized, the material is

(A) n-type with carrier concentration of 10^{16}/cm^3

(B) p-type with carrier concentration of 10^{16}/cm^3

(C) p-type with carrier concentration of 2 ×10^{16}/cm^3

(D) n-type with carrier concentration of 2 ×10^{16}/cm^3

Open solution

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Q56. GATE ECE 1987

In an intrinsic semiconductor the free electron concentration depends on

(A) Effective mass of electrons only.

(B) Effective mass of holes only.

(C) Temperature of the Semiconductor.

(D) Width of the forbidden energy band of the semiconductor.

Open solution

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Answer Key

Marks 1

1. **A, D**	2. **A**	3. **A, B, C**	4. **A**	5. **A**
6. **A**	7. **C**	8. **A**	9. **A**	10. **A**
11. **14**	12. **1.92**	13. **0.52**	14. **A**	15. **1.13**
16. **D**	17. **13**	18. **C**	19. **A**	20. **A**
21. **B**	22. **A**	23. **C**	24. **C**	25. **C**
26. **D**	27. **B**	28. **A**	29. **B**	30. **B**
31. **D**	32. **C**	33. **B**	34. **C**	

Marks 2

35. **(35.47) (34.50 to 36.50)**	36. **B**	37. **B**	38. **0.17 to 0.19**	39. **2.20 to 2.30**
40. **69.87**	41. **A, C**	42. **B**	43. **C**	44. **1.6**
45. **100**	46. **225.2**	47. **A**	48. **4000**	49. **C**
50. **A**	51. **A**	52. **D**	53. **A**	54. **A**
55. **B**	56. **C, D**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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